

MAGNITOGORSK AP, Russian Federation

WMO: 232215

Lat: 53.393N

Lon: 58.756E

Elev: 1430

StdP: 13.95

Time Zone: 5.00 (E05)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.3	-14.7	-24.0	1.5	-17.5	-20.4	1.9	-14.1	25.5	11.4	23.2	12.0	4.2	10	0.674

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	87.7	63.2	84.2	63.1	80.8	61.6	66.9	80.4	65.3	78.7	63.8	76.3	7.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.4	89.0	71.4	60.8	83.9	69.6	59.1	79.0	68.5	32.2	79.2	31.0	78.5	29.9	76.0	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4)	24.1	21.2	18.9	DB	-22.8	91.8	5.3	2.7	-26.6	93.7	-29.7	95.3	-32.6	96.8	-36.4	98.7
(5)				WB	-22.7	70.4	5.0	1.7	-26.3	71.7	-29.2	72.7	-32.1	73.7	-35.7	74.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	3.8	7.5	20.3	41.2	55.6	63.9	66.3	63.9	52.5	38.4	24.7	11.2
	DBStd	24.15	10.62	11.61	9.18	9.36	8.72	7.24	6.07	7.79	7.32	8.97	10.95	11.58
	HDD50	6273	1432	1190	920	295	44	3	0	3	54	369	760	1203
	HDD65	10276	1897	1610	1385	715	308	106	56	117	381	823	1210	1668
	CDD50	1746	0	0	0	30	218	420	506	434	127	11	0	0
	CDD65	275	0	0	0	0	17	73	97	83	5	0	0	0
	CDH74	3085	0	0	0	9	295	839	1003	877	62	0	0	0
	CDH80	881	0	0	0	0	38	246	280	308	9	0	0	0
Wind	WSAvg	8.8	7.6	7.9	9.8	10.1	9.5	9.0	8.3	8.2	8.1	9.4	9.3	8.2
Precipitation	PrecAvg	23.5	1.8	1.4	1.8	1.5	1.9	2.5	2.9	2.3	1.7	1.9	1.9	2.0
	PrecMax	32.2	3.3	3.1	3.5	3.7	3.6	5.4	6.1	6.2	3.3	4.4	3.5	5.0
	PrecMin	17.9	0.3	0.2	0.1	0.2	0.2	0.6	0.8	1.0	0.5	0.4	0.2	0.5
	PrecStd	4.0	0.7	0.7	0.9	0.8	0.9	1.1	1.3	1.1	0.7	0.9	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.0	36.0	46.0	75.3	84.0	89.8	91.4	91.3	80.6	66.4	48.5	35.4
		MCWB	28.5	31.3	36.4	52.7	59.9	61.4	63.0	64.8	59.3	49.2	43.5	32.3
	2%	DB	27.0	32.2	37.8	68.3	80.5	86.2	86.3	87.6	73.8	60.7	44.2	31.7
		MCWB	25.5	29.4	32.1	50.8	58.1	62.2	63.9	64.3	56.0	48.7	40.1	29.6
	5%	DB	24.6	28.4	35.5	62.7	77.3	82.5	82.6	83.9	69.9	55.2	40.9	29.8
		MCWB	22.9	26.2	31.6	48.5	56.4	61.2	63.0	64.1	55.3	44.9	37.5	28.1
	10%	DB	20.8	23.3	33.4	57.3	73.6	78.9	79.2	78.8	65.8	50.4	37.4	26.9
		MCWB	19.1	21.1	30.7	46.1	54.8	60.6	62.0	63.1	53.5	43.8	34.4	24.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.0	32.9	36.7	55.1	62.1	67.6	68.3	68.5	61.7	52.3	44.8	31.6
		MCDB	29.2	34.9	43.3	70.3	78.4	80.9	81.0	81.5	75.9	62.3	48.2	33.0
	2%	WB	25.8	29.4	33.3	51.8	59.7	65.2	66.8	66.9	58.6	48.9	40.5	29.9
		MCDB	27.1	31.6	35.4	65.8	76.5	78.6	80.6	80.5	70.5	57.7	42.8	31.2
	5%	WB	22.5	26.0	32.0	49.5	58.0	63.5	65.2	65.3	56.6	46.8	37.6	27.2
		MCDB	24.2	28.5	34.6	60.7	73.2	76.6	78.3	79.0	66.2	53.6	40.0	29.0
	10%	WB	18.7	21.5	30.2	46.6	56.0	62.0	63.7	63.7	54.6	44.6	34.5	25.3
		MCDB	20.5	23.5	33.0	56.8	70.5	74.5	75.3	76.7	63.2	49.5	36.7	27.0
Mean Daily Temperature Range	5% DB	MDBR	12.7	14.9	14.6	18.3	23.5	21.9	20.9	20.1	18.4	14.0	11.3	12.1
		MCDBR	11.5	15.5	15.5	27.1	27.6	27.0	26.5	27.3	25.0	22.3	13.3	12.0
		MCWBR	10.7	12.9	12.2	13.7	11.5	10.4	9.6	9.6	11.3	12.5	9.8	10.9
	5% WB	MCDBR	11.2	14.4	14.5	24.9	25.8	22.3	22.4	23.7	21.6	18.4	11.3	11.6
		MCWBR	10.6	12.1	12.4	13.8	11.5	9.7	9.5	9.6	11.1	11.5	9.3	10.7
Clear-Sky Solar Irradiance	taub	0.272	0.298	0.352	0.401	0.391	0.385	0.406	0.422	0.367	0.335	0.299	0.268	
	taud	2.126	2.096	2.005	2.123	2.274	2.360	2.297	2.229	2.349	2.334	2.234	2.122	
	Ebn at Noon	216	249	257	261	272	275	266	252	254	236	206	192	
	Edh at Noon	22	31	43	43	40	37	38	39	30	25	20	18	
All-Sky Solar Radiation	RadAvg	331	677	1155	1449	1736	1897	1788	1460	1015	552	279	219	
	RadStd	28	48	88	127	150	161	120	141	102	95	36	29	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page